

AI-Enhanced Curriculum Design and Deep-Learning-Based Assessment in International Sports Communication Education

Fangni Li

 <https://orcid.org/0000-0003-3356-1640>

Communication University of China, China

Received: July 10th, 2025 | **Accepted:** October 1st, 2025

ABSTRACT

Traditional assessment in international sports communication is often fragmented and subjective, limiting timely, learner-centered feedback. This study presents a curriculum framework enhanced by generative artificial intelligence, coupled with a deep learning (DL) model for instructional effectiveness assessment in international sports communication. The pipeline integrates de-identified learning analytics—learning management system clickstreams, interaction networks, and rubric-scored artifacts—into engineered features for DL training with parameter search and cross-validation. A 16-week field study across three undergraduate sections at a Chinese comprehensive university (N = 60; two involving generative artificial intelligence, one comparison) benchmarked DL against linear regression and decision tree baselines and against expert ratings on intercultural communication competence, framing diversity, and production quality. Results show that DL converged faster and yielded lower prediction error than the baselines, while closely aligning with expert scores, enabling actionable, personalized feedback and course tuning driven by constructive alignment.

KEYWORDS

Generative Artificial Intelligence, International Sports Communication, Curriculum System, Instructional Effect, Assessment Model

INTRODUCTION

In the 21st century, the wave of science and technology is bringing generative artificial intelligence (GenAI) into ubiquitous use throughout everyday life, offering a unique charm and unlimited potential (Xu et al., 2021; Saam, 2024). In the realm of education, GenAI has introduced revolutionary possibilities, such as personalized learning, intelligent content delivery, and data-driven instructional methods, thereby reshaping traditional pedagogical practices (Wang et al., 2024). Nevertheless, they have yet to be fully explored in terms of interdisciplinary domains, such as international sports communication, which is the convergence of studies on sports and communication theory, as well as global cultural exchange (Holmes et al., 2019; Wei et al., 2020). In today's connected world, international sports communication is increasingly becoming an important matter. The other purpose is to spread sports-related information and promote international cultural understanding and cooperation.

DOI: 10.4018/IJICTE.391357

This article published as an Open Access article distributed under the terms of the Creative Commons Attribution License (<http://creativecommons.org/licenses/by/4.0/>) which permits unrestricted use, distribution, and production in any medium, provided the author of the original work and original publication source are properly credited.

In an era marked by sports diplomacy, this field is crucial (Logo, 2018; Murray & Pigman, 2014). Although significant, these challenges are inherent to the current emphasis on curriculum and instruction in this domain. In traditional approaches, there is a lack of adaptability to global trends and the exploitation of cutting-edge technologies, such as GenAI, in enhancing teaching and learning outcomes. Therefore, the relatively limited development of graduates with innovative capabilities and international perspectives, which would make them competent to thrive in an evolving global context, is hindered (Dugalić, 2018; Murray & Pigman, 2014; Oluyemisi, 2023).

The intricacy of integrating athletics, communication, and international relations results in fragmented curricula that fail to provide cohesive learning experiences (Räikkönen & Hedman, 2025). Furthermore, standard assessment methods, such as standardized tests and subjective feedback, are insufficient for evaluating the varied range of skills necessary for this interdisciplinary work (Virk et al., 2020). The lack of such systematic, data-driven evaluation frameworks hinders instructional progress and the quantification of educational impact. GenAI is an advanced technical means for simulating intelligent human behavior, rapid data processing, precise information dissemination, and extensive knowledge mining (Alfredo et al., 2024; Jain & Raghuram, 2024). GenAI is also useful for helping teachers modify course content and train new teaching strategies. It offers more personalized and efficient learning to students in the international sports communication course (Bhutoria, 2022). In this study, we aim to explore a new approach to developing an international sports communication curriculum system and evaluating its instructional effectiveness using GenAI. This paper aims to analyze the potential application and real-world use of GenAI technology in the international sports communication curriculum system from several dimensions, including curriculum objectives setting, curriculum content design, instructional method innovation, and assessment system construction. Concurrently, a deep learning (DL) model is implemented and scientifically assessed to analyze the instructional effect of GenAI in the field of international sports communication education.

This paper addresses these pressing issues through a proposal for a comprehensive curriculum system for international sports communication that features GenAI-driven instructional support and assessment. First, this study contributes dually by enriching the theoretical and practical foundations of international sports communication education via the integration of GenAI into curriculum design and assessment. It also offers a scalable, data-driven framework to improve instruction quality, realize personalized learning, and graduate with global and innovative competencies. This research ultimately enhances international sports communication education by modernizing and supporting learning environments with the use of GenAI in higher education. This study bridges the gap between technological innovation and pedagogical practice, providing important insights into constructing more adaptive and inclusive education systems.

THEORETICAL FRAMEWORK

This research integrates four complementary strands.

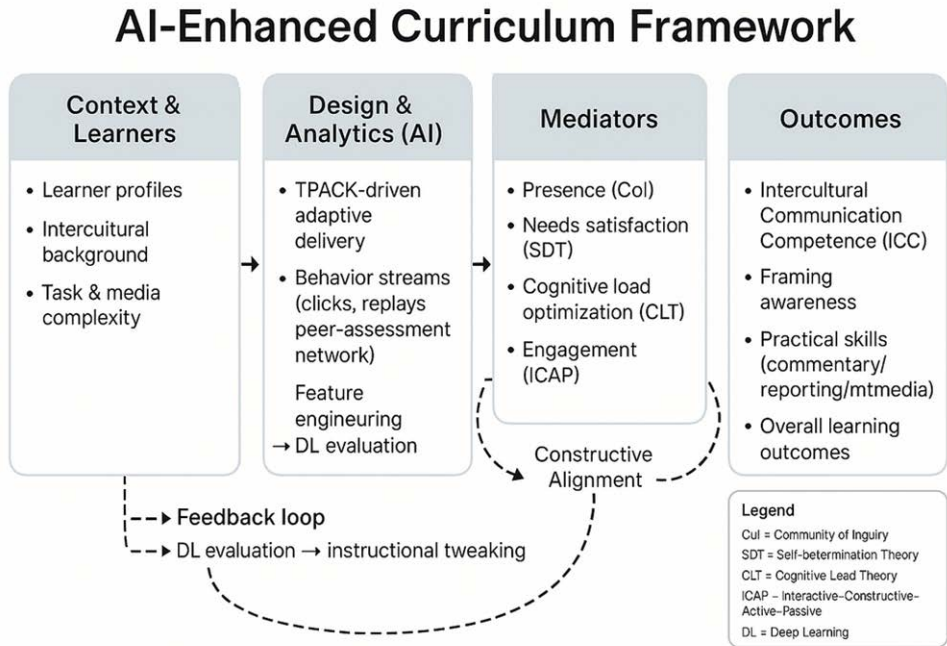
- **Constructive alignment and Technological Pedagogical Content Knowledge framework (TPACK):** justify aligning intended outcomes (e.g., intercultural reporting, on-air commentary) with learning activities and assessments and position GenAI as a design mediator across content, pedagogy, and technology (Biggs, 1996; Mishra & Koehler, 2006).
- **Community of inquiry (teaching, social, cognitive presence) and self-determination theory (autonomy, competence, relatedness):** explain how adaptive feedback, learner choice, and peer interaction increase engagement and persistence (Deci & Ryan, 2000; Garrison et al., 2000).
- **Cognitive load theory and Interactive–Constructive–Active–Passive (ICAP) framework:** inform multimedia/interaction design and predict that optimized load and higher-order engagement (interactive/constructive) yield superior learning (Chi & Wylie, 2014; Sweller, 1988; Sweller et al., 1998).

- **intercultural communication competence (ICC) and framing theory:** for discipline-specific outcomes, define what good performance means in international sports communication—cultural perspective-taking, appropriate language use, frame diversity, and evidence-based storytelling (Byram, 1997; Deardorff, 2006; Entman, 1993).

Figure 1 outlines the various assessment dimensions and corresponding methods used to evaluate students' performance in the course. It includes assessments on knowledge mastery, skill application, literacy enhancement, and learning attitude.

The framework links inputs (context and learners) to design and analytics (GenAI), where constructive alignment and TPACK justify the alignment of intended outcomes, learning activities, and assessments and motivate the use of GenAI as a design mediator. Behavioral streams (such as clicks, replays, and peer-assessment networks) are transformed into features for deep learning evaluation. Effects are transmitted through learning mechanisms: Community of Inquiry (CoI) framework (teaching, social, and cognitive presence), Self-Determination Theory (SDT) (support for autonomy, competence, and relatedness), Cognitive Load Theory (CLT) (optimized cognitive load via segmentation and signaling), and the Interactive–Constructive–Active–Passive (ICAP) framework (shift from passive or active learning to constructive and interactive engagement). Outcomes are discipline-specific, encompassing Intercultural Communication Competence (ICC), framing awareness, and practical production and commentary skills, which are aggregated into overall achievement. The feedback loop utilizes model outputs (e.g., presence index, autonomy score, ICAP index, load proxies) to prioritize instructional adjustments, thereby enhancing the alignment between outcomes, activities, and assessments. This figure specifies testable relations (e.g., positive effects of presence/autonomy; an inverted-U between load proxies and outcomes) that are operationalized in the Method section.

Figure 1. Mechanism of the GenAI-Enhanced Curriculum for International Sports Communication.



Note. Col = Community of Inquiry; SDT = Self-Determination Theory; CLT = Cognitive Load Theory; ICAP = Interactive-Constructive-Active-Passive framework; ICC = Intercultural Communication Competence; TPACK = Technological Pedagogical Content Knowledge; DL = Deep Learning; GenAI = Generative Artificial Intelligence.

CONSTRUCTION OF AN INTERNATIONAL SPORTS COMMUNICATION CURRICULUM SYSTEM

Under the backdrop of globalization, the importance of international sports communication is increasingly prominent, as it is not only related to the transmission of sports information but also serves as an important bridge between cultural exchange and international understanding (Onwumechili, 2017). Therefore, it is particularly important to construct a scientific, comprehensive, and international sports communication curriculum system that meets the needs of the times.

Curriculum Goal Setting

The goal of an international sports communication course should focus on three dimensions: knowledge, skills, and accomplishment, aiming to cultivate students with a solid theoretical foundation, practical skills, a broad international perspective, and cross-cultural communication skills. The specific objectives are shown in Table 1. It outlines the various assessment dimensions and corresponding methods used to evaluate students' performance in the course, including assessments of knowledge mastery, skill application, literacy enhancement, and learning attitude.

Table 1. Course Objectives

Dimension	Specific objectives
Knowledge	Master basic theories, development history, and current status of international sports communication; understand international sports events, rules, and culture; familiarize with international communication theories and practices.
Skills	Possess abilities in information collection, editing, and publishing; master multimedia production and dissemination techniques; capable of cross-cultural communication and collaboration.
Literacy	Possess international perspective and cross-cultural awareness; possess strong critical thinking and innovation abilities; establish correct sports values and communication ethics.

Course Content Design

The design of the international sports communication course content should be systematic, practical, and employ foresight to ensure that students can fully master the knowledge and skills in this field. The course content can be roughly divided into several modules as shown in Table 2. It provides an overview of various modules and their associated course examples in the field of sports studies. It highlights key subjects ranging from the history of international sports to practical skills in sports reporting, multimedia production, and cross-cultural communication. Each module is designed to equip students with specialized knowledge and skills essential for working in the global sports industry.

Table 2. Course Content Design

Module name	Example course titles
International sports foundation	History of International Sports, Overview of Global Sports Events
Sports communication theory and practice	Introduction to Sports Communication, Cross-Cultural Sports Communication
Multimedia production and dissemination techniques	Sports News Photography and Videography, Sports Video Editing
Sports event reporting and commentary	Practical Sports Event Reporting, Sports Commentary Writing
International sports organizations & rules	International Olympic Committee and International Sports Federations, International Sports Law
Cross-cultural communication and collaboration	Intercultural Communication Studies, Sports Team Management and Collaboration

In order to enhance the practicality and interactivity of the course, we can also set up teaching links such as case analysis, simulation drills, and field visits, such as organizing students to report, plan, and simulate live broadcast of international large-scale sports events, to improve students' actual combat ability (Heinrichs, 2008).

Innovation of Instructional Methods

In teaching an international sports communication course, we should actively explore and adopt various instructional methods to meet the diverse learning needs and styles of different students. The following are some suggested instructional methods:

- **teaching method:** It is suitable for imparting theoretical knowledge, but attention should be paid to increasing the interaction between teachers and students to avoid full-time irrigation.
- **discussion method:** Encourage students to actively participate in classroom discussions and cultivate critical thinking and expression skills, which can be carried out around hot topics or cases.
- **case analysis:** Analyzing real cases helps students to understand the application of theoretical knowledge in practice, which is suitable for sports communication practice courses.
- **simulation exercise method:** By simulating real scenes, students' practical ability and teamwork ability can be improved, such as simulating sports event reports.
- **on-the-spot investigation method:** Organize students to visit stadiums, media organizations, etc., to enhance perceptual knowledge and practical experience.
- **online learning method:** Use Internet resources to carry out online learning, broaden learning channels, and increase time flexibility.

Construction of Assessment System

Constructing a scientific and reasonable assessment system is an important means to ensure teaching quality and learning effect. The assessment system for international sports communication courses should encompass two aspects: process assessment and summative assessment and employ a diversified range of assessment methods. Table 3 introduces the basic dimensions and methods of the assessment system. It outlines the assessment dimensions and corresponding methods used to evaluate students' performance in the course. It categorizes assessments into four key areas: knowledge mastery, skill application, literacy enhancement, and learning attitude. These methods include exams, assignments, discussions, and participation, ensuring a comprehensive evaluation of students' academic and practical abilities.

Table 3. Assessment System

Assessment dimension	Assessment methods
Knowledge mastery	Midterm and final exams, course papers
Skill application	Practical assignments, simulation exercises
Literacy enhancement	Class discussions, group collaborations
Learning attitude	Attendance, homework submission

In the concrete implementation, the assessment methods and weights can be appropriately adjusted according to the course characteristics and teaching objectives. At the same time, a feedback mechanism should be established to collect feedback from students and teachers in a timely manner, allowing for continuous optimization of the assessment system.

METHOD

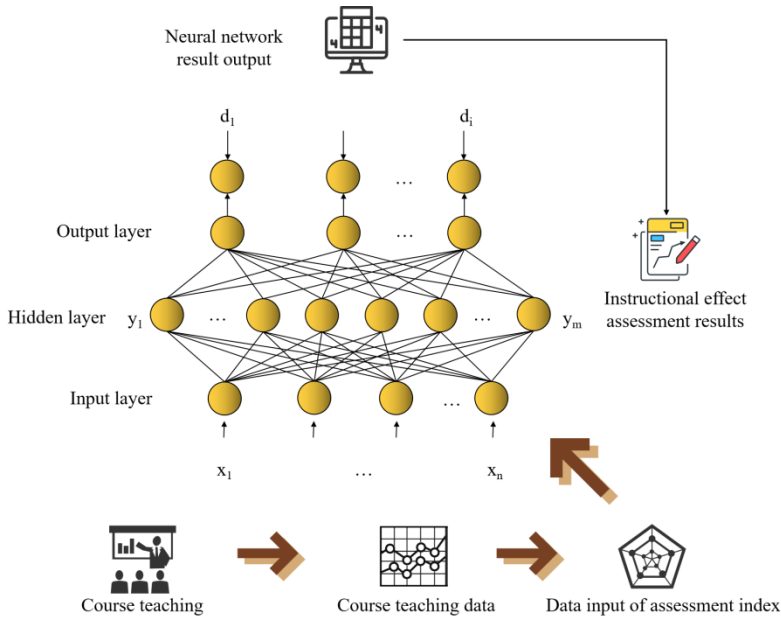
Traditional methods of assessing instructional effectiveness relied on subjective measures such as test scores and questionnaires, which made it challenging to comprehensively and objectively reflect students' learning status and teachers' instructional performance (Meylani, 2024; Wood & Doğan, 2024). With the advancement of educational informatization, large volumes of data, including learning behavior data and interaction data, were generated during the teaching process. The data contained rich information that required deep exploration and utilization. Instructional effect assessment models

based on DL leveraged these extensive datasets and complex algorithmic models to automatically uncover patterns in the teaching process, providing a scientific and objective foundation for evaluating instructional effectiveness (Cui et al., 2023; Li & Zhang, 2024).

As a branch of machine learning, the core principle of DL involves simulating the learning processes of the human brain through the construction of deep neural network models, enabling nonlinear mapping and feature extraction of complex data. In the context of instructional effect assessment, DL technology was applied to several key areas (Taye, 2023). First, it facilitated the recognition of students' behavioral patterns by analyzing their learning behavior data, thereby identifying distinct learning styles and habits. Second, it enabled the prediction of knowledge mastery by using students' historical learning data and performance information to forecast their future understanding of the subject matter. Finally, it supported the generation of feedback on instructional effectiveness, offering personalized teaching recommendations to educators based on the assessment results.

The instructional effect assessment model of international sports communication courses based on DL is shown in Figure 2. It illustrates the end-to-end learning analytics pipeline that supports AI-enhanced instruction. It integrates classroom interactions, learning management system activity, and assignment artifacts (gold nodes), transforms them via feature engineering and network analysis, and outputs dashboards and competency profiles to guide targeted feedback and course refinement.

Figure 2. Instructional Effect Assessment Model



Let $x_i(t)$ denote the input information at time t and $o_j(t)$ denote the output information at time j at t . Then, the state of neuron j is represented as:

$$o_j(t) = f\left\{\left[\sum_{i=1}^n \omega_{ij} x_i(t - \tau_{ij})\right] - T_j\right\} \quad (1)$$

with τ_{ij} as the synaptic delay, T_j as the neuron threshold, ω_{ij} representing the weight from i to j of neurons, and $f(\bullet)$ as the transfer function. If τ_{ij} order corresponds to unit time, then:

$$o_j(t+1) = f\left\{\left[\sum_{i=1}^n \omega_{ij} x_i(t)\right] - T_j\right\} \quad (2)$$

The subscripts of input and output signify the diversity of modes in the neuron model, enabling us to leverage this model property to address various problems based on different requirements. The neuronal input at time t is formulated as follows:

$$net_j(t) = \sum_{i=1}^n \omega_{ij} x_i(t) \quad (3)$$

Regarding the aforementioned formula, neurons are effective solely when $net_j(t) > T_j$. Upon simplification, the neuron model transforms into:

$$o_j = f(net_j) = f(W_j^T X) \quad (4)$$

First, various types of data were collected during the teaching process. These included students' learning behavior data (such as click streams, learning duration, and homework completion), interactive data (such as classroom discussions and online question-and-answer sessions), and exam results. These data were then cleaned, denoised, and standardized to ensure their quality and usability. From the preprocessed data, features relevant to instructional effect assessment were extracted. These features included aspects such as students' study habits (e.g., study frequency and study periods), knowledge mastery (e.g., accuracy rates and error types), and interaction metrics (e.g., discussion activity and question quality). Feature selection techniques were applied to identify the key features that significantly impacted the assessment results.

The DL algorithm was then utilized to construct the assessment model, and the extracted feature data were used to train the model. This study draws on de-identified learning analytics and coursework artifacts generated during the routine delivery of three undergraduate sections of International Sports Communication at one Chinese comprehensive university. Data collection took place over 16 weeks in 2025. Two sections (GenAI-enabled) integrated the proposed analytics workflow, while one matched comparison section followed business-as-usual instruction without analytics. The analytic sample comprised $N = 60$ students (GenAI-enabled: $n = 40$; comparison: $n = 20$; 57% female; mean age 19.9 ± 1.1 years). All records were captured as part of normal teaching operations; the researchers accessed no personally identifying information.

This study ingested three streams:

1. **behavioral traces from the learning management system and media server:** clickstream events, session-level dwell time, replay-pause counts, back-scrolls, and submission latency—78,960 timestamped events in total (median ~1,300 per student across the term; video plays = 18,360; replays = 6,150)
2. **interaction traces from course forums and peer assessment:** original posts, replies, and @-mentions—yielding 540 posts, 1,180 replies, and 145 @-mentions. These supported the construction of reply networks (density, reciprocity, clustering).
3. **assessment artifacts:** graded exams and rubric-assessed coursework, 158 student artifacts in total: reporting pieces ($n = 66$), commentary scripts ($n = 56$), and multimedia packages ($n = 36$).

To benchmark model outputs, five domain experts (with at least 8 years of experience in teaching or industry in sports journalism or broadcasting) independently rated a stratified sample of 120 artifacts using validated rubrics for ICC, framing diversity, and practical production quality. A 20% overlapping subset (24 artifacts) was double-marked to monitor inter-rater reliability; disagreements were resolved by moderated discussion prior to aggregation. During the training process, the accuracy and generalization ability of the model were enhanced by adjusting model parameters and optimizing the algorithms. Additionally, the stability and reliability of the model were evaluated through cross-validation. Once each index assessment was completed according to the assessment system's criteria, the original index data matrix was standardized and transformed to finalize the entire instructional effect assessment process. Given m samples for assessment and n assessment indexes, the original data matrix undergoes standardization and transformation as follows.

$$\bar{x}_j = \frac{\sum_{i=1}^m x_{ij}}{m} \quad (5)$$

$$s_j = \sqrt{\frac{1}{m-1} \sum_{i=1}^m (x_{ij} - \bar{x}_j)^2} \quad (6)$$

where $\bar{x}_j$ denotes the sample average of the j index and s_j represents the sample standard deviation of the j index. Following the standardized transformation, each sample of data exhibits a mean of 0 and a variance of 1. By considering the variables in the assessment model, the membership vector of each index is determined through statistical methods, with reference to the quantitative indicators of the teaching assessment index system for the international sports communication curriculum.

To mitigate overfitting during model training, the initial maximum depth of each regression tree is set to a small value. The size of the updating step determines the convergence speed of the model. In this article, three parameters, denoted as M , S , α , are selected through parameter enumeration and assessed using cross-validation. The combination with the highest assessment score is chosen as the final model parameter value, with the assessment standard being goodness of fit:

$$R^2 = 1 - \frac{\sum_{i=1}^m (Y_i - y_i)^2}{\sum_{i=1}^m (y_i - y')^2} \quad (7)$$

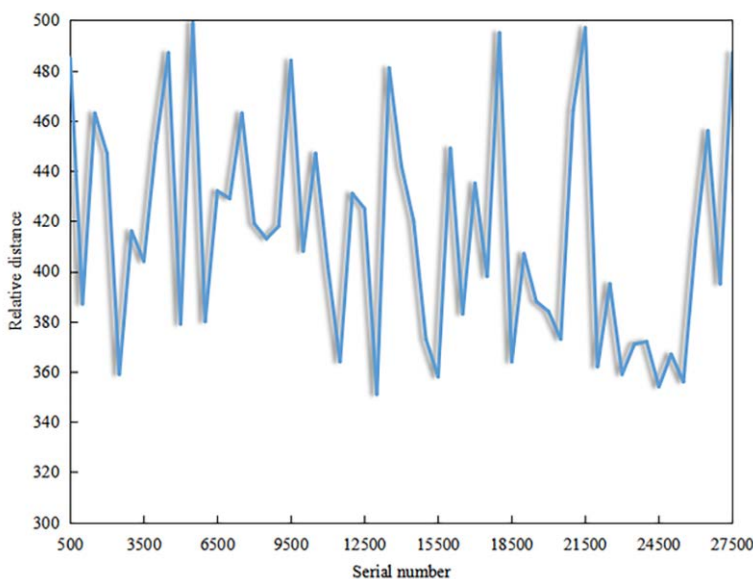
where R^2 represents goodness of fit, m denotes the number of samples, Y_i is the actual value of samples, y_i is the predicted value of samples, y' signifies the average value of prediction, and a value closer to 1 indicates a better fitting effect.

RESULTS

When preparing the data for the instructional effect assessment model, it is crucial to perform thorough preprocessing to ensure the data remains clean and consistent. A necessary preprocessing step was removing outlying observations, or extreme values, where observations had values much greater or lower than those typically observed in the dataset. Outlying observations may occur for several reasons, such as measurement error, data entry error, or other irregular conditions at the time of data collection. If outliers are not addressed, they will skew the data and change the training of the model by not being able to learn the “true” underlying relationships and patterns, which leads to model overfitting and limits the model's ability to generalize, and lower predictive accuracy. For this study, a statistical procedure was employed to identify and remove outlying observations, ensuring the dataset could be used to train the model without outliers causing abnormal data values.

Figure 3 illustrates the data before and after removing outlying observations to understand better how removing outliers can graphically affect the data. In particular, the chart shows that removing the outlier leads to a decrease in data fluctuation, which is important in itself for trusting the dataset and its use in future analyses. This cannot be overstated in importance. This figure plots relative distance against the serial number (indexed observations), visualizing fluctuation patterns across the dataset. It is used to assess dispersion and stability, and to flag potential anomalies. Sharp peaks indicate large relative distances, while troughs indicate proximity. The wide oscillations suggest substantial heterogeneity across samples, motivating the use of subsequent smoothing, robust modeling, and outlier checks before analysis.

Figure 3. Data Outlier Removal Processing

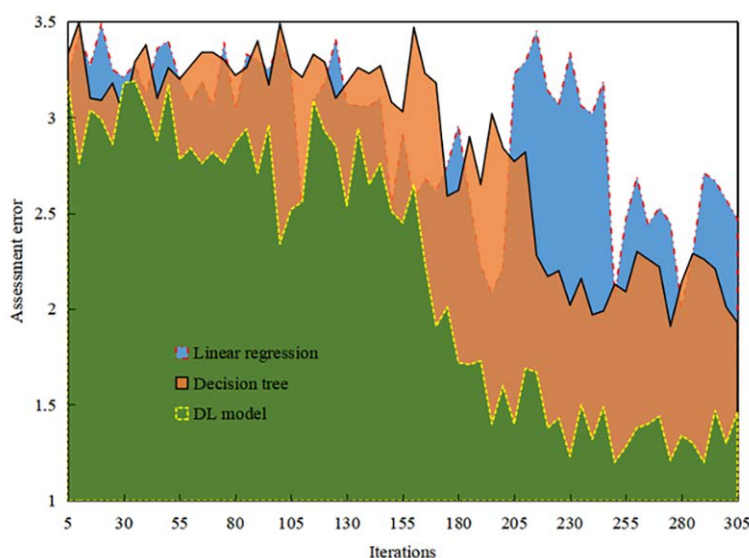


In the instructional effect assessment model training procedure, the preprocessed data is the first and most essential step. The DL algorithm applied to the model selects the most complex patterns to capture and learn, which are nonlinear and intricate in the dataset. The execution phase of the training process involves adjusting the network weights to iterate, with the aim of the model gradually progressing toward the correct way to identify the relationships and characteristics inherent in the data. This iterative refinement process that imprints the DL model is such that the feature enabling the DL models to meander through is the difficulties of the dataset over the course of training cycles. As a consequence of this focus, the DL model is thoroughly tested for performance by comparing it against a variety of algorithms. The specific algorithms, including the traditional linear regression model and the decision tree model, are chosen for a benchmark comparison and are selected as the default ones. Although these models are simpler in their design, they are frequently used for predictive analytics and provide a baseline against which the DL model is estimated.

The comparison involves calculating the prediction errors for each algorithm when it is used on the same dataset, thus enabling an objective evaluation of their performance relative to one another. The results of these comparative experiments, as shown in Figure 4, are presented in the form of a visual representation of the models' assessment errors across multiple iterations. The figure illustrates that the DL model (the green line with a yellow dotted pattern) all the time shows lower

assessment errors in relation to the linear regression model (the blue line) and the decision tree model (the orange line). As the iteration number increases, the error rate for the DL model continues to decrease, indicating its superior learning performance and effective dataset adaptation. On the other hand, the linear regression and decision tree models generally exhibit higher and more fluctuating error rates, especially in the initial iterations, which is a consequence of their slow convergence and inability to capture the inherent complexity of the data effectively. The gradual improvement in the DL model's performance has demonstrated its robustness and efficiency in the instructional effect assessment problem.

Figure 4. Error Reduction Across Iterations (Supports CLT-Informed Segmentation Hypothesis)



Note. CLT = Cognitive Load Theory; DL = Deep Learning.

The visual representation effectively demonstrates that DL techniques have an edge over traditional algorithms in predictive accuracy. The DL model's consistent decline in assessment error, observed across iterations, has been further supported by the iterative optimization process, which has effectively enhanced the model's performance. This comparative analysis of the performance of the two models will provide the most credible confirmation that the DL model is superior in dealing with complex data, which will, in turn, demonstrate its capacity for delivering accurate and reliable assessments of the effects of instruction.

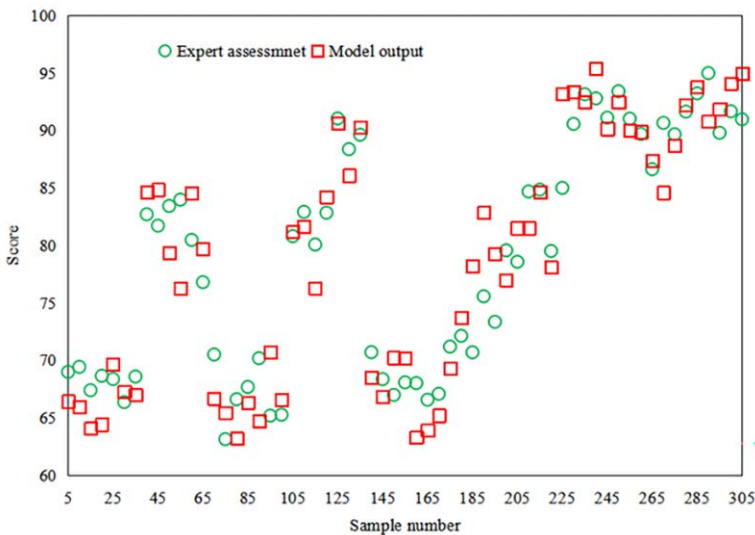
Figure 4 compares assessment error across 305 iterations for three models (linear regression, decision tree, and a DL model). Lower is better: the DL model shows the most consistent decline and stabilizes near ~1.2 after ~200 iterations, outperforming the decision tree (gradual drop to ~2.0) and linear regression (more volatile with higher errors), indicating the DL model delivers the best overall performance.

Contrary to traditional assessment models, the instructional effect assessment model introduced in this study, which is based on the DL model, is a distinct alternative, particularly because it has the capacity for faster convergence and lower prediction errors. This means that the DL model is a more efficient way to discover the data's underlying issues and thus to propose a better explanation for its instructional effectiveness. Additionally, the model is tested for its correctness and real-world applicability in a simulation experiment specifically designed for this purpose. In this

experiment, the training set is formed by selecting a subset of the total sample of data, and the trained DL model is then used to generate predictions for the rest of the data. These prediction outcomes are compared to the actual assessment results provided by the domain experts, and the model's performance is quantified against expert evaluations in a direct comparison. Within this framework of analysis, the research aims to assess the predictive reliability of the model and its applicability in authentic educational settings. The simulation experiment results provide decisive proof for the DL model developed through simulation-expert relations, demonstrating its efficiency in imitating expert judgments, which serve as the basis for proposing the method to supplant or complement traditional teaching outcome assessment. Within this methodology, the comparison of the model's actual outputs with experts' knowledge base predictions provides strong evidence of the model's robustness and its capacity for predictions driven by reliable data. Thus, the model's practical value in educational assessment is further proved.

Figure 5 compares expert assessments (green circles) with model outputs (red squares) across 305 samples. The model tracks the experts' trend and ranking well—especially in later samples ($\approx 220\text{--}305$)—with small, systematic deviations: slight underestimation at lower-score clusters ($\approx 62\text{--}70$) and mild overestimation near local peaks ($\approx 85\text{--}92$). The close co-movement indicates strong alignment with expert scoring and supports using the model for scalable assessment.

Figure 5. Expert Assessment Results and Model Output Results



The output of the model in Figure 5, alongside the expert assessments' results, facilitates a detailed comparison that allows for a direct evaluation of the model's performance in predicting instructional effects. The figure features the green circles to indicate the assessment scores allocated by domain experts, while the red squares are the corresponding output of the trained model. The presented plot clearly indicates that the model's predictions compare favorably with the majority of cases, as evaluated by domain experts, and the output scores show a high degree of consistency with the expert scores. Such a situation can be interpreted as the model's ability to effectively cope with the expert's decisions, thereby accurately predicting the instructional methods. Additionally, the resemblance of the presentation of the two sets of results, particularly the general distribution of scores, indicates the model's high-level precision in capturing the data's underlying patterns.

The distance between the model's predictions and the experts' ratings is particularly notable, as it provides empirical support for the model's accuracy and predictive power. The findings also provide reassurance that the DL-based instructional effect assessment model we proposed is not only capable of generating predictions that closely resemble expert ratings but may also have utility for real-world applications. This provides further support for the model's utility in educational assessments, particularly in terms of its ability to produce trustworthy and predictable outcomes. The model's predictions visually resemble those of the expert assessment in Figure 4, affirming both practical, real-world, and theoretical feasibility.

DISCUSSION

This research found that higher social presence and autonomy scores jointly predict performance, aligning with the Community of Inquiry (CoI) framework and Self-Determination Theory (SDT) accounts of engagement. The observed inverted-U relation between cognitive-load proxies and outcomes supports CLT-based task design. Moreover, improvements in framing diversity and ICC rubric scores explain discipline-specific gains in international sports reporting, validating the choice of framing theory and ICC as outcome lenses. Together with constructive alignment and TPACK, these results suggest that GenAI functions not merely as a tool but as a design mediator that tightens the outcome–activity–assessment loop.

The findings of this study demonstrate how the use of GenAI, specifically DL, has the potential to transform instructional design and assessment as a part of the academic environment for international sports communication. This research integrates GenAI into curriculum development and instructional effect assessment to address the long-standing challenges addressed in prior studies and introduce novel solutions. This study developed an instructional effect assessment model based on the constructivist learning theory, in which personalized learning experiences and instructor-controlled instruction strategies play a major role. Putting the constructivist construct that learning occurs best in an individualized way to the test, this model successfully processes diverse datasets from students' learning behavior and patterns of interaction. Like previous studies, this study has found that GenAI can be used to enhance personalized education. For example, Agarwal and Agarwal (2024) utilized neural networks in higher education, achieving improved predictive accuracy of instructional outcomes in a more limited setting. This study presents more applications than these works, as the integration of GenAI is designed within an interdisciplinary curriculum framework and is particularly dedicated to the dynamic and complex nature of international sports communication.

The study's findings align with earlier research, which highlights the limitations of conventional assessment methods based on subjective evaluations and static scores (Abdulrahman et al., 2020; Heydarnejad et al., 2022). This study overcomes these limitations by employing DL algorithms to obtain a more objective, scalable, and nuanced assessment model. In Figure 3, for both centralized and decentralized selection functions, the proposed DL model is seen to converge faster than traditional methods such as linear regression and the decision tree model. Consistent with Gameil and Al-Abdullatif (2023), these advantages imply that multimedia and data-driven platforms can optimize instructional design. However, unlike previous research, the present research utilizes DL to assess unidimensional instructional effects based on behavioral, interactive, and academic performance data (Elahi et al., 2023). This approach is demonstrated to be robust and reliable, as evidenced by the close agreement between the DL model's predictions, as shown in Figure 4, and expert assessments. This is also in accordance with Zhang (2023), who proposed integrating wireless network technology into teaching assessment, but without a particular emphasis on GenAI or sophisticated algorithms. This study extends the technological integration in education by employing DL, providing precise feedback for teachers and students that should lead to increased instructional quality (Keiper et al., 2023; Ouyang & Jiao, 2021).

The design of the GenAI-assisted curriculum system addresses the gaps identified by Kasztelnik (2024), who highlighted the obstacles to cohesively integrating interdisciplinary content into traditional curricula. Since the curriculum framework requires the inclusion of theoretical knowledge, practical skills, and cultural literacy, it ensures a holistic learning experience and runs against Uljens and Ylimaki (2017) critique of fragmented curricula in sports education. In addition, GenAI supporting the choice of curriculum evokes a change in the direction of a more flexible and living educational model that becomes necessary for the current international sports communication. Figure 5 illustrates the data preprocessing techniques employed to address the common issue of outliers in educational datasets. The removal of the anomalies facilitates the assessment of the instructor effect model. De-kun and Memon (2022) also discussed others that relied on simpler statistical models; hence, similar approaches were designed for mobile evaluation algorithms in physical education. In this study, we bring DL one step closer to practical application by adopting it for a more sophisticated treatment of complex data and better accuracy in prediction and feedback capability.

The practical implications of the findings go beyond purely theoretical contributions. Through an instructional effect assessment model that delivers the personalized feedback sought by Janelli and Lipnevich (2021), this study addresses the practical difficulties documented by Al Shloul et al. (2024) regarding the need for innovative instruments to evaluate and enhance instructional quality. This study's model is scalable and transferable to diverse educational contexts, both globally and across interdisciplinary fields. The work presented here overlaps with previous research in some ways, but it does so in important and distinct ways. For instance, Alam and Mohanty (2023) employed principal component analysis for evaluating physical education, focusing on ideational integration rather than instructional design and assessment. On the other hand, this study proposes a holistic curriculum framework that relies extensively on advanced GenAI techniques. As Ospankulov et al. (2023) did with the study of international sports talent training, his work does not reach the technological depth achieved here, especially the use of DL for instructional assessment.

Generally, by focusing on GenAI-based innovations, this study aligns with an emerging trend in school reform, as noted by Wang et al. (2024). More specifically, this research departs from the conventional setting in which GenAI is regarded as an instrument for assessment and instead makes an original contribution by recognizing that it can also serve as a forum for curriculum innovation (Pinski et al., 2024; Walter, 2024). This study demonstrates the advantages of DL models, including rapid convergence and minimal prediction error (Figure 3), thereby giving actionable insights to educators and policymakers interested in modernizing interdisciplinary education. However, this study has some limitations despite its many contributions. For model training and validation, the data primarily from a single educational context may not be generalized. Furthermore, the DL model accurately predicts instructional effects but encounters scalability issues in data-poor settings, due to its reliance on high-quality data.

CONCLUSION

This study explored the construction of an international sports communication curriculum system and an instructional effect assessment framework, leveraging AI, specifically DL algorithms. This research provides innovative solutions to address gaps in traditional curriculum systems and assessment methods, thereby enhancing the effectiveness, adaptability, and inclusivity of international sports communication education. This study concludes by providing a robust framework for integrating artificial intelligence (AI) into international sports communication education. It invents a new pathway bridging the divide between conventional pedagogy and technical advancement, leading the way for a richer, more inclusive, and better educational ground. In a world where AI technology is evolving, the integration of AI into education will bring new possibilities to learning and teaching in various disciplines. Therefore, the research also makes substantial contributions to the theory and practice of education as it explores ways to combine AI into curriculum design and assessment processes. A

comprehensive curriculum system that combines theoretical knowledge, practical skills, and cultural literacy addresses a large gap in the current literature. This responds to the critique by Zhao et al. (2019) regarding the fragmentation of curricula in sports communication education and offers a cohesive framework that aligns with globalization and interdisciplinary learning. Consequently, the proposed DL-based assessment model advances educational assessment beyond the limitations of subjective assessment to provide precise, data-driven insights into students' learning behavior and instructional effectiveness.

This research fills a gap in that there is a lack of studies that harness AI-driven methodologies with interdisciplinary curriculum development work. This study differed from previous studies in that it used an integrated approach to applying AI to both curriculum design and instructional effect assessment. This work extends previous research by demonstrating that, in complex interdisciplinary settings such as international sports communication, AI tools can be utilized effectively, thereby expanding the role AI plays in education. Implications of this study range from the academic to the practical. It academically highlights the ability of AI to profoundly alter the foundations of traditional pedagogy and introduce a model for future interdisciplinary education with emergent technologies. The research is actionable, enabling educators and policymakers to provide adaptive, data-driven curricula that accommodate a broad range of learner needs while simultaneously improving instructional outcomes.

However, the study has its limitations. Model training and validation were conducted using a dataset from a specific educational context, which limits the generalizability of the findings. Additionally, while the DL-based model can predict instructional outcomes, its effective application is significantly impacted by its reliance on high-quality, well-structured data in resource-limited or data-poor settings. The second limitation is the lack of ethical consideration, which includes the omission of data privacy and security. The future of AI-assisted educational models may involve real-time feedback mechanisms and the integration of adaptive learning platforms into the models. Secondly, cooperation between educators, technologists, and policymakers on ethics, technology, and pedagogy is necessary to address the ethical, technological, and pedagogical challenges of integrating AI-driven solutions. Future research could address these limitations by enhancing the model's generalizability by incorporating data from diverse educational contexts and disciplines into the expanded dataset. Furthermore, hybrid AI is considered for its use with various approaches and DL methods to improve adaptability in data-sparse contexts. Additionally, reviewing ethical frameworks to facilitate responsible AI use in education and protect the use of student data. Finally, conducting longitudinal studies on the long-term influence of implementing AI-enriched curricula on student achievement, with implications for the workforce, becomes paramount.

COMPETING INTERESTS

The authors of this publication declare there are no competing interests.

FUNDING

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. Funding for this research was covered by the author(s) of the article.

CORRESPONDING AUTHOR

Correspondence should be addressed to Fangni Li; lifangni@cuc.edu.cn.

REFERENCES

- Abdulrahman, M. D., Faruk, N., Oloyede, A. A., Surajudeen-Bakinde, N. T., Olawoyin, L. A., Mejabi, O. V., & Azeez, A. L. (2020). Multimedia tools in the teaching and learning processes: A systematic review. *Heliyon*, 6(11), e05312. DOI: 10.1016/j.heliyon.2020.e05312
- Agarwal, M., & Agarwal, B. B. (2024). Predicting student academic performance using neural networks: Analyzing the impact of transfer functions, momentum and learning rate. *International Journal of Experimental Research and Review*, 40(Spl Volume), 56–72. DOI: 10.52756/ijerr.2024.v40spl.005
- Al Shloul, T., Mazhar, T., Abbas, Q., Iqbal, M., Ghadi, Y. Y., Shahzad, T., & Hamam, H. (2024). Role of activity-based learning and ChatGPT on students' performance in education. *Computers and Education: Artificial Intelligence*, 6, 100219. DOI: 10.1016/j.caeai.2024.100219
- Alam, A., & Mohanty, A. (2023). Educational technology: Exploring the convergence of technology and pedagogy through mobility, interactivity, AI, and learning tools. *Cogent Engineering*, 10(2), 2283282. DOI: 10.1080/23311916.2023.2283282
- Alfredo, R., Echeverria, V., Jin, Y., Yan, L., Swiecki, Z., Gašević, D., & Martinez-Maldonado, R. (2024). Human-centred learning analytics and AI in education: A systematic literature review. *Computers and Education: Artificial Intelligence*, 5, 100215. DOI: 10.1016/j.caeai.2024.100215
- Bhutoria, A. (2022). Personalized education and artificial intelligence in the United States, China, and India: A systematic review using a human-in-the-loop model. *Computers and Education: Artificial Intelligence*, 3, 100068. DOI: 10.1016/j.caeai.2022.100068
- Biggs, J. (1996). Enhancing teaching through constructive alignment. *Higher Education*, 32(3), 347–364. DOI: 10.1007/BF00138871
- Byram, M. (1997). *Teaching and assessing intercultural communicative competence*. Multilingual Matters. DOI: 10.2307/jj.22730614.6
- Chi, M. T. H., & Wylie, R. (2014). The ICAP framework: Linking cognitive engagement to active learning outcomes. *Educational Psychologist*, 49(4), 219–243. DOI: 10.1080/00461520.2014.965823
- Cui, Y., Ma, Z., Wang, L., Yang, A., Liu, Q., Kong, S., & Wang, H. (2023). A survey on big data-enabled innovative online education systems during the COVID-19 pandemic. *Journal of Innovation & Knowledge*, 8(1), 100295. DOI: 10.1016/j.jik.2022.100295
- De-kun, J., & Memon, F. H. (2022). Design of mobile intelligent evaluation algorithm in physical education teaching. *Mobile Networks and Applications*, 27(2), 527–534. DOI: 10.1007/s11036-021-01818-1
- Deardorff, D. K. (2006). Identification and assessment of intercultural competence as a student outcome of internationalization. *Journal of Studies in International Education*, 10(3), 241–266. DOI: 10.1177/1028315306287002
- Deci, E. L., & Ryan, R. M. (2000). The “what” and “why” of goal pursuits: Human needs and the self-determination of behavior. *Psychological Inquiry*, 11(4), 227–268. DOI: 10.1207/S15327965PLI1104_01
- Dugalić, S. (2018). Sport, media and digitalization. *Sport–Science & Practice*, 8(1), 56–69.
- Elahi, M., Afolaranmi, S. O., Martinez Lastra, J. L., & Perez Garcia, J. A. (2023). A comprehensive literature review of the applications of AI techniques through the lifecycle of industrial equipment. *Discover Artificial Intelligence*, 3(1), 43. DOI: 10.1007/s44163-023-00089-x
- Entman, R. M. (1993). Framing: Toward clarification of a fractured paradigm. *Journal of Communication*, 43(4), 51–58. DOI: 10.1111/j.1460-2466.1993.tb01304.x
- Gameil, A. A., & Al-Abdullatif, A. M. (2023). Using digital learning platforms to enhance the instructional design competencies and learning engagement of preservice teachers. *Education Sciences*, 13(4), 334. DOI: 10.3390/educsci13040334
- Garrison, D. R., Anderson, T., & Archer, W. (2000). Critical inquiry in a text-based environment: Computer conferencing in higher education. *The Internet and Higher Education*, 2(2–3), 87–105. DOI: 10.1016/S1096-7516(00)00016-6

- Heinrichs, W. L., Youngblood, P., Harter, P. M., & Dev, P. (2008). Simulation for team training and assessment: Case studies of online training with virtual worlds. *World Journal of Surgery*, 32(2), 161–170. DOI: 10.1007/s00268-007-9354-2
- Heydarnejad, T., Tagavipour, F., Patra, I., & Farid Khafaga, A. (2022). The impacts of performance-based assessment on reading comprehension achievement, academic motivation, foreign language anxiety, and students' self-efficacy. *Language Testing in Asia*, 12(1), 51. DOI: 10.1186/s40468-022-00202-4
- Holmes, W., Bialik, M., & Fadel, C. (2019). *Artificial intelligence in education promises and implications for teaching and learning*. Center for Curriculum Redesign. DOI: 10.25215/9358094575.02
- Jain, K., & Raghuram, J. N. V. (2024). Unlocking potential: The impact of AI on education technology. *Multidisciplinary Reviews*, 7(3), 2024049–2024049. DOI: 10.31893/multirev.2024049
- Janelli, M., & Lipnevich, A. A. (2021). Effects of pre-tests and feedback on performance outcomes and persistence in massive open online courses. *Computers & Education*, 161, 104076. DOI: 10.1016/j.compedu.2020.104076
- Kasztelnik, K. (2024). Artificial intelligence-assisted curriculum development: Innovations in designing educational content for the 21st century learner. *Journal of Higher Education Theory and Practice*, 24(11), 51–59. DOI: 10.33423/jhetp.v24i11.7367
- Keiper, M. C., Fried, G., Lupinek, J., & Nordstrom, H. (2023). Artificial intelligence in sport management education: Playing the AI game with ChatGPT. *Journal of Hospitality, Leisure, Sport and Tourism Education*, 33, 100456. DOI: 10.1016/j.jhlste.2023.100456
- Li, Y., & Zhang, H. (2024). Big data technology for teaching quality monitoring and improvement in higher education: Joint *k*-means clustering algorithm and Apriori algorithm. *Systems and Soft Computing*, 6, 200125. DOI: 10.1016/j.sasc.2024.200125
- Logo, E. D. (2018). *The role of sports diplomacy in contemporary international relations: A case study of the Olympic Games* (Doctoral dissertation, University of Ghana). DOI: 10.28942/ssj.v6i3.801
- Meylani, R. (2024). A comparative analysis of traditional and modern approaches to assessment and evaluation in education. *Bati Anadolu Eğitim Bilimleri Dergisi*, 15(1), 520–555. DOI: 10.51460/baebd.1386737
- Mishra, P., & Koehler, M. J. (2006). Technological pedagogical content knowledge: A framework for teacher knowledge. *Teachers College Record*, 108(6), 1017–1054. DOI: 10.1111/j.1467-9620.2006.00684.x
- Murray, S., & Pigman, G. A. (2014). Mapping the relationship between international sport and diplomacy. *Sport in Society*, 17(9), 1098–1118. DOI: 10.1080/17430437.2013.856616
- Oluyemisi, O. M. (2023). Impact of artificial intelligence in curriculum development in Nigerian tertiary education. *International Journal of Educational Research*, 12(2), 192–211. DOI: 10.70382/tijerls.v07i8.023
- Onwumechili, C. (2017). *Sport communication: An international approach*. Routledge. DOI: 10.4324/9781315270920
- Ospankulov, Y., Zhumabayeva, A., Nishanbayeva, S., Ussen, B., & Zhalel, A. (2023). The effect of the use of digital technologies in physical education lessons on students' physical education cultures and attitudes towards the lesson. *International Journal of Education in Mathematics. Science and Technology*, 11(6), 1424–1442. DOI: 10.46328/ijemst.3700
- Ouyang, F., & Jiao, P. (2021). Artificial intelligence in education: The three paradigms. *Computers and Education: Artificial Intelligence*, 2, 100020. DOI: 10.1016/j.caeai.2021.100020
- Pinski, M., & Benlian, A. (2024). AI literacy for users: A comprehensive review and future research directions of learning methods, components, and effects. *Computers in Human Behavior: Artificial Humans*, 1(1), 100062. DOI: 10.1016/j.chbah.2024.100062
- Räikkönen, T., & Hedman, J. (2025). Unlocking the power of sports: An exploration of the nexus between shared place, community competence, and sense of community. *International Journal of Sport Policy*, 17(1), 81–99. DOI: 10.1080/19406940.2024.2396836
- Saam, M. (2024). The impact of artificial intelligence on productivity and employment: How can we assess it and what can we observe? *Inter Economics*, 59(1), 22–27. DOI: 10.2478/ie-2024-0006

- Sweller, J. (1988). Cognitive load during problem solving: Effects on learning. *Cognitive Science*, 12(2), 257–285. DOI: 10.1207/s15516709cog1202_4
- Sweller, J., van Merriënboer, J. J. G., & Paas, F. G. W. C. (1998). Cognitive architecture and instructional design. *Educational Psychology Review*, 10(3), 251–296. DOI: 10.1023/A:1022193728205
- Taye, M. M. (2023). Understanding of machine learning with deep learning: Architectures, workflow, applications and future directions. *Computers*, 12(5), 91. DOI: 10.3390/computers12050091
- Uljens, M., & Ylimaki, R. M. (2017). Bridging educational leadership, curriculum theory and Didaktik: Non-affirmative theory of education. In Uljens, M., & Ylimaki, R. M. (Eds.), *Bridging educational leadership, curriculum theory and Didaktik* (pp. 3–145). Springer. DOI: 10.1007/978-3-319-58650-2_1
- Virk, A., Joshi, A., Mahajan, R., & Singh, T. (2020). The power of subjectivity in competency-based assessment. *Journal of Postgraduate Medicine*, 66(4), 200–205. DOI: 10.4103/jpgm.JPGM_591_20
- Walter, Y. (2024). Embracing the future of Artificial Intelligence in the classroom: The relevance of AI literacy, prompt engineering, and critical thinking in modern education. *International Journal of Educational Technology in Higher Education*, 21(1), 15. DOI: 10.1186/s41239-024-00448-3
- Wang, S., Wang, F., Zhu, Z., Wang, J., Tran, T., & Du, Z. (2024). Artificial intelligence in education: A systematic literature review. *Expert Systems with Applications*, 252, 124167. DOI: 10.1016/j.eswa.2024.124167
- Wei, Y., Zhang, H., & Yin, H. (2020). Development of communication and sport in China based on mapping of knowledge information visualization. *SAGE Open*, 10(1). DOI: 10.1177/2158244019900175
- Wood, P. C., & Doğan, C. D. (2024). A comparative study on assessment methods used by high school teachers in Uganda and Turkey. *Research in Comparative and International Education*, 19(3), 352–370. DOI: 10.1177/17454999241256447
- Xu, Y., Liu, X., Cao, X., Huang, C., Liu, E., Qian, S., & Zhang, J. (2021). Artificial intelligence: A powerful paradigm for scientific research. *Innovation (Cambridge (Mass.))*, 2(4), 100179. DOI: 10.1016/j.xinn.2021.100179
- Zhang, J. (2023). School wireless network classroom teaching system based on artificial intelligence. *Applied Artificial Intelligence*, 37(1), 2219563. DOI: 10.1080/08839514.2023.2219563

Li Fangni, Ph.D. in Sports Humanities and Sociology, Born in Kunming, Yunnan, China. She is a Lecturer at the School of International Studies, Communication University of China. With 12 years of experience in both Chinese and international media, she has worked as a host and reporter for media outlets such as China Central Television (CCTV), Xinhua News Agency, and Fox Sports. She has over 3 million followers across all social media platforms, with her videos accumulating more than 100 million views in total.